Questions for Practice

Quiz 1 (Week1-4)

(selected questions from AQ, PA, GA and SWI)

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Note: Solutions provided are may vary, are provided to best of my knowledge

Week1:

Q1) Which of the following is true about machine learning?

- ☐ It is a study of algorithms that improve through the use of data.
- ☐ It is a sub field of AI that extracts patterns out of raw data.
- ☐ It allows computers to learn from experience without being explicitly programmed or human intervention.
- ☐ It is used for tasks where programming/human labour fails.
- ☐ All of these

Q2) Mark all incorrect statements in the following

- ☐ 1(355%2 = 1) = 1
- ☐ 1(788%2 = 1) = 0
- ☐ 1(355%2 = 0) = 1
- ☐ 1(788%2 = 0) = 1

Q3) Which of the following is false regarding supervised and unsupervised machine learning?

a. Unsupervised machine learning helps you to find different kinds of unknown patterns in data.
b. Regression and classification are two types of supervised machine learning techniques while clustering and density estimation are two types of unsupervised learning.
c. In unsupervised learning model, the data contains both input and output variables while in supervised learning model, the data contains only input data

Q4) Consider the following data set where each data point consists of three

x_1	x_2	x_3
10	10	9
13	12	13
5	5	4
8	7	7

features x_1, x_2, x_3

Consider two encoder functions f and $\bar{f}$ with decoders g and $\bar{g}$ respectively aiming to reduce the dimensionality of the data set from 3 to 1:

Pair 1: $f(x_1, x_2, x_3) = x_1 - x_2 + x_3$ and $g(u) = [u, u, u]$

Pair 2: $\bar{f}(x_1, x_2, x_3) = \frac{x_1 + x_2 + x_3}{3}$ and $\bar{g}(u) = [u, u, u]$

The reconstruction loss of the encoder decoder pair is the mean of the squared distance between the actual input and reconstructed input.

a. Loss for pair 1: ________________
b. Loss for pair 2: ________________

Q5) Consider an encoder **W** and decoder **W**^{T} functions given below:

$$\mathbf{W} = \begin{bmatrix} 1,0,0,0 \\ 0,1,0,0 \end{bmatrix}$$

Compress a data point **x** = $[1,2,3,4]^T$ to obtain **u** and reconstruct **x'** from **u.** How close is the reconstruction to the original?______________________

Q6) $f(x_1, x_2, x_3) = (x_1+2\ x_2)/3$ is used as encoder function and $g(u)=[u,2u,3u]$ is used as decoder function for dimensionality reduction of following data set.

X
[1,2,3]
[2,3,4]
[-1,0,1]
[0,1,1]

Give the reconstruction error for this encoder decoder pair. The reconstruction error is the mean of the squared distance between actual input and reconstructed input.__________________

Q7) The encoder and decoder functions expressed as

$f(x) = x_1 - x_2 + 0.5x_3$ and $g(u) = [0.3u, -u, u]$

to reduce the dimension of the data set given below:

x
$[-1, 1, -1]$
$[1, 2, 1]$
$[1, 1, -1]$
$[1, -1, 2]$
$[0, 1, 3]$

What is the loss obtained for the model? _____________________

Q8) Consider the following input data points:

$\mathbf{X}$	y
$[4, 2]$	$+1$
$[8, 4]$	$+1$
$[2, 6]$	-1
$[4, 10]$	-1
$[10, 2]$	$+1$
$[12, 8]$	-1

What will be the average misclassification error when the functions $g(X) = sign(x_1 - x_2 - 2)$ and $h(X) = sign(x_1 + x_2 - 10)$ are used to classify the data points into classes $+1$ or -1.

a. Loss for pair g: __________________
b. Loss for pair h: __________________

Q9)The table below shows the original labels and predicted labels (classes) of some **Multiclass** classification problems. Compute the squared error loss and 0-1 loss. Which loss function seems to be a good one?

Labels/Ground Truth	**Predicted**
1	1
2	4
3	1
4	4
1	4
2	2
3	3
4	1
1	1
1	1
1	1

a. *Squared error loss*
b. *0-1 Loss*

Q11) Consider the following data set

x	y
$[1, 0]$	$+1$
$[2, 2]$	$+1$
$[3, 1]$	$+1$
$[4, 5]$	-1
$[5, 4]$	-1
$[6, 5]$	-1

When a function that outputs the sign(x_1 - $3x_2$) of is used as a classification model to fit the data set, what is the resultant loss?________________

Q12) Consider the following 4 training examples. We want to learn a function

x	y
-1	0.0319
0	0.8692
1	1.9566
2	3.0343

$f(x)=ax+b$ which is parameterized by (a,b). Using average squared error as the loss function, which of the following parameters would be best to model the given data?

a. (1, 1)
b. (1, 2)
c. (2, 1)
d. (2, 2)

Q13) Consider the following input data points:

X	y
[2]	5.8
[3]	8.3
[6]	18.3
[7]	21
[8]	22

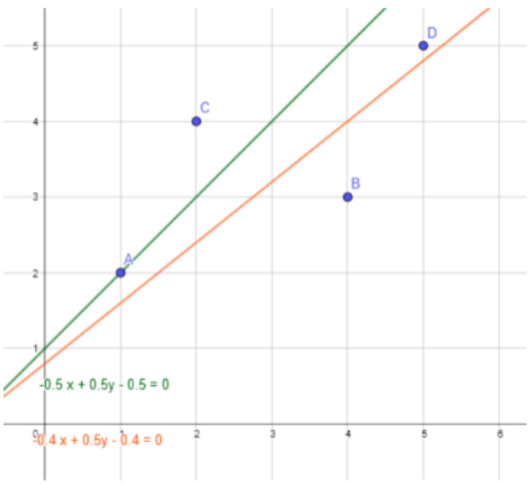
What will be the amount of loss when the functions $g = 3x_1 + 1$ $h = 2x_1 + 2$ are used to represent the regression line. Consider the average squared error as a loss function.

a. Loss for pair g: ________________
b. Loss for pair h: ________________

Q14) The mean squared error loss obtained while fitting this function on the data set is________________

For the data set $(x^i, y^i) = [(1,2),(3,4),(4,3),(5,6)]$, i= 1 to 4, let the regression function be $\frac{x^2 - 2y}{5}$.

Q15) Look at the graph below and answer the following questions



Which of these regression lines is the best one (in the MSE sense)

a. y = x+1
b. $y = 0.8x+0.8$

Q16) Consider a line with zero slope and y-intercept of 2. The sum of squared residuals obtained when this line is used to fit the following data set is ______________

x	y
1	2
2	4
3	1
4	3

Q17) P(X) is obtained by the density estimation algorithm for the data points $x_1=2.5$, $x_2=1$, $x_3=3$, $x_4=4.5$, $x_5=4.95$

A probability model $P(X) = \begin{cases} \frac{1}{5}, & \text{if } x \in [0,5] \\ 0, & \text{otherwise} \end{cases}$

Compute the negative log likelihood loss of the model.__________________

Solutions:

1. All of these

2. A,b,d
3. C
4. 3,0.66
5. 25
6. 8.63
7. 4.628
8. 0.166, 4.98
9. B
10. -
11. 0.166
12. A
13. 2.964, 11.924
14. 8 to 9.5
15. B
16. 6
17. 0.698

Week - 2 (AQs)

Q1) If the vectors u and v belong to a vector space V, then,

A. their linear combination belongs to V.
B. their linear combination is perpendicular to V.
C. their linear combination is zero.
D. their linear combination may belong to any vector space other than V.

Q2) For a real-valued function, which of the following is true?

A. Domain is R
B. Co-domain is R
C. Both domain and co-domain are R

Q3) Choose the correct statement/s from the following:

A. If a function is continuous at a point, then it is also differentiable at that point.
B. If a function is differentiable at a point, then it is also continuous at that point.
C. If a function is not continuous at a point, then it is not differentiable at that point.
D. If a function is not differentiable at a point, then it may or may not be continuous at that point.

Q4) The plot of linear approximation of a function ***f*** at a point **v** is

A. perpendicular to **f**
B. parallel to **f**
C. tangent to ***f*** at the point **(v,f(v))**

Q5) Which of the following vectors is/are perpendicular to any given vector?

A. [1 1 1]
B. [0 1 1]
C. [1 0 1]
D. [0 0 0]
E. [-1 0 1]

Q6) Which of the following represents a multivariate function?

A. $f:\mathbb{R}\rightarrow\mathbb{R}$
B. $f:\mathbb{R}^d\rightarrow\mathbb{R}$
C. $f:\mathbb{R}\rightarrow\mathbb{R}^d$
D. $f:\mathbb{R}\rightarrow\mathbb{R}\times\mathbb{R}$

Q7) Contour plots of linear functions look like

A. equidistantly spaced circles.
B. non-uniformly spaced circles.
C. equidistantly spaced straight lines.
D. non-uniformly spaced straight lines.

Q8) Which of the following statements are true?

(i) Gradient of a function points in the direction of the steepest descent.
(ii) Gradient of a function points in the direction of the steepest ascent.
(iii) Gradient of a function at a point is parallel to the contour through the point.
(iv) Gradient of a function at a point is perpendicular to the contour through the point.

A. (i) and (iii)
B. (ii) and (iv)
C. (ii) and (iii)
D. (i) and (iv)

Q9) The approximation of $f(x_1, x_2) = x_1^2 + x_2^2$ around (3,1) is given by

A. $6x_2 - 2x_1 - 10$
B. $6x_2 + 2x_1 - 10$
C. $6x_2 - 2x_1 + 10$
D. $6x_2 + 2x_1 + 10$

Q10) For the function $f(x) = x^2 - x$

A. The critical point represents maxima.
B. The critical point represents minima.
C. The critical point is a saddle point.

<u>Solutions:</u>

1. A
2. B
3. B,c,d
4. C
5. D
6. B
7. C
8. B
9. B
10. b

Week - 2 (PAs)
Q1) *sin*(*x*)/*x* is continuous at x=0

A. True
B. False

Q2) Consider two d-dimensional vectors *x* and *y* and the following terms:

(i) $x^T y$ (ii) x.y (iii) $\sum x_i y_i$

Which of the above terms are equivalent?

A. Only (i) and (ii)
B. Only (ii) and (iii)
C. Only (i) and (iii)
D. (i), (ii) and (iii)

Q3) The linear approximation of tan(x)

A. 1 + x
B. 1 - x
C. x
D. 1-x

Q4) Consider the following function, is f(x) continuous?

$$f(x) = \begin{cases} 7x + 2, & \text{if } x > 1 \\ 9, & \text{if } x \leq 1 \end{cases}$$

A. True
B. False

Q5) Which of the following is the best approximation of e^{0.019} (Use linear approximation around 0).

A. 1
B. 0
C. 0.019
D. 1.019

Q6) What is the linear approximation of $f(x,y) = x^2 + y^2$ around(1,1)?

A. 2x + 2y +2
B. 2x + 2y -2
C. 2x - 2y - 2
D. 2x + 2y +1

Q7) The directional derivative of $f(x,y,z)=x^2+3y+z^2$ at (1, 2, 1) along the unit vector in the direction of [1, -2, 1] is (correct upto three decimal places)__________________________

Q8) Find the direction of steepest ascent for the function $x^2+y^3+z^4$ at point (1, 1, 1) ______________ ?

Q9) The directional derivative of f[x,y,z]=x+y+z at point [-1, 1, -1] along the unit vector in the direction of [1, -1, 1] is (correct upto three decimal places)

____

Q10) The cubic approximation of $(1+x)^{0.5}$ at a = 0 is _____________

Solutions:

1. B
2. D
3. C
4. A
5. D
6. B
7. -0.856 to -0.775
8. 1/(sqrt(29) [2 3 4]
9. 0.548 to 0.605
10. 1 + x/2 -⅛ (x^2) +1/16(x^3)

Week 2: (GAs)

Q1) Which of the following functions is/are continuous?

A. 1/(x-1)
B. $(x^2-1)/(x-1)$
C. sign(x-2)
D. sin(x)

Q2) Regarding a *d*-dimensional vector *x*, which of the following four options is not equivalent to the rest two options?

A. x^Tx
B. $||x^2||$
C. xx^T

Q3) For two vectors a and b, which of the following is true as per Cauchy-Schwarz inequality?

(i) $a^T b \leq ||a|| * ||b||$
(ii) $a^T b \geq -||a|| * ||b||$
(iii) $a^T b \geq ||a|| * ||b||$
(iv) $a^T b \leq -||a|| * ||b||$

A. (i) only
B. (ii) only
C. (iii) only
D. (iv) only
E. (i) and (ii)
F. (iii) and (iv)

Q4) Which of the following is the equation of the line passing through [7,8,6] in the direction of vector [1, 2, 3]?

A. $[1,2,3]+\alpha[-6,-6,3]$
B. $[7,8,9]+\alpha[-6,-6,3]$
C. $[1,2,3]+\alpha[6,6,3]$
D. $[7,8,6]+\alpha[6,6,3]$
E. $[7,8,6]+\alpha[1,2,3]$
F. $[1,2,3]+\alpha[7,8,6]$

Q5)____________________________________ ?

Approximate the value of $e^{0.011}$ by linearizing e^x around x=0.

Q6) ____ ?

Approximate $\sqrt{3.9}$ by linearizing $\sqrt{x}$ around $x = 4$.

Q7) The directional derivative of $f(x,y,z)=x^3+y^2+z^3$ at (1, 1, 1) in the direction of unit vector along v=[1, -2, 1] is (correct upto three decimal places)________________

Q8) The direction of steepest ascent for the function $2x+y^3+4z$ at the point (1, 0, 1) is ____________

Q9) For $f(x, y) = x^2y$

A. Derivative in the direction of (1, 2) at the point (3, 2) is ____________
B. The derivative of f in the direction of (2, 1) at the same point, i. e., (3, 2). Is ____________
C. In which direction is the derivative maximum? ____________
D. What is the value of the maximal derivative? ____________
E. What is the derivative in the direction of (-3, 4) at the point (3, 2)? ____________
F. What is the derivative in the direction of (-4, -3) at the point (3, 2)? ____________

Q10) Without computing, tell whether x.y = ||x|| ||y|| , if x = [2, 5, 6] and y = [16, 40, 48]

A. Yes
B. No

Solutions:

1. D
2. C
3. E
4. E
5. 1.011
6. 1.975
7. 0.775 to 0.856
8. 1/sqrt(20) [2 0 4]
9. 13.41, 14.75, (2,1), (12/15, 9/15), 0
10. A

Week -2: Others

Q1) Consider the function f(x) shown below:

$$f(x) = \begin{cases} x^3 & \text{if } x > 1, \\ x^2 & \text{if } 0 < x \leq 1 \\ x & \text{if } x < 0, \\ 0 & \text{if } x = 0 \end{cases}$$

A. *f* is continuous, but not differentiable at x=1
B. *f* is both continuous and differentiable at x=1
C. *f* is continuous, but not differentiable at x=0
D. *f* is not continuous at x=0
E. *f* is not continuous at x=1
F. *f* is not continuous at x=1
G. *f* is both continuous and differentiable at x=0

Q2) Which of the following options are correct?

$$f(x) = \begin{cases} -x^2 + 2x + 3 & \text{if } 0 \leq x \leq 50 \\ x^3 + 3 & \text{if } -50 \leq x < 0. \end{cases}$$

A. 1 is a local maximum.
B. −50 is the global minimum.
C. 0 is the global maximum.
D. 50 is the global minimum.

Q3) Consider f(x) = x^3 - x, answer which of the following is/are correct?

A. *f* has neither local maxima nor local minima.
B. sqrt{2}is a local minimum.

C. sqrt{2}is a local maximum.
D. -sqrt{2}is a local maximum.
E. -sqrt{2} is a local minimum.
F. *f* has two critical points.

Q4) Find out the directional derivative of the function $f(x,y)=x^2y^3$ at (1,2), in the direction of the vector (0,2) ________________________

Q5) $L_f(x,y)$ = ______________________________

$$f(x,y) = ye^x - \tfrac{1}{4}(x^2 + y^2) \text{ at } (0,1).$$

Q6) Determine the equation of the tangent line at x=0.5 and using it estimate the value of $g(1.5)$ __________

$$g(x) = 2.5e^{-x^2+0.2x+2}.$$

Q7) The directional derivative of $f(x,y)=xy^2+3x^2$ at (1,1) in the direction of unit vector along w=[-1,1] is ___________ ?

Q8) For the function below at (1,1) the first element and second element of the gradient of g(x,y) are ___________ &__________________

$$g(x,y) = \sqrt{2x^2 - y^3}$$

Q9) A function $f(x,y)=x^2+2xy^3$ is approximated linearly in the neighbourhood of (2,-2). Use the approximation to approximate f(2.3,-2.2) ____________ ?

Q10) Find all point where G is not continuous:

$$\text{Let } G(x) = \begin{cases} \frac{1}{(x+3)^2} & , \text{ if } x \le -1; \\ 2 - x & , \text{ if } -1 < x \le 1; \\ \frac{3}{x+2} & , \text{ if } x > 1. \end{cases}$$

<u>Solutions:</u>

1. A,c
2. A,b
3. B,d,e
4. 12
5. x+y/2+¼
6. 3.51
7. Range -4,-3
8. Range 1,2; range -0.6, -0.5
9. -41.2
10. -3

Week 3: (AQs)

Q1) Gaussian elimination changes the null space of a matrix.

A. True
B. False

Q2) If a $n \times n$ matrix A is invertible, then which of the following is/are false?

A. C(A) is R^n
B. $Ax=b$ has infinite solutions.
C. $Ax=b$ has a unique solution.
D. N(A) only contains zero vector.

Q3) Which of the following is/are true?

A. $C(A) \perp N(A)$
B. $R(A) \perp N(A)$
C. $C(A) \perp N(A^T)$
D. $R(A) \perp N(A^T)$

Q4) A system of equations $Ax=b$ is inconsistent if:

A. $b \in R(A)$
B. $b \notin R(A)$
C. $b \in C(A)$
D. $b \notin C(A)$

Q5) The column space of the projection matrix of a vector v is

A. a line parallel to the line through v
B. a line passing through v
C. a line perpendicular to the line through v

Q6) Consider the following statements regarding a projection matrix P

1. $P^2=P$
2. P is symmetric
3. $P^2 \neq P$
4. P is not symmetric

Which of the above statements are true?

A. 1 only
B. 1 and 2
C. 2 only
D. 1, 2 and 3
E. 3 and 4

Q7) The null space of the projection matrix of a vector v is

A. a line parallel to the line through v.
B. a line passing through v.
C. a line perpendicular to the line through v.
D. a plane orthogonal to v.

Q8) Consider Ax=b The projection matrix of A is an identity matrix if

A. $b \in C(A)$
B. $b \in R(A)$
C. $b \in N(A)$
D. $b \in N(A^T)$

Q9) Consider Ax=b The projection matrix of A is an zero matrix if

A. $b \in C(A)$
B. $b \in R(A)$
C. $b \in N(A)$
D. $b \in N(A^T)$

Q10) If $A\theta=b$ has a solution then,

A. $||A\theta-b||^2 = 1$
B. $||A\theta-b||^2 = 0$
C. $||A\theta-b||^2 < 1$
D. $||A\theta-b||^2 > 1$

Solutions:

1. B
2. B
3. B,c
4. D
5. B
6. B
7. C
8. A
9. D
10. b

Week 3: (PAs)

Q1) Rank of matrix is ________________

$$\begin{bmatrix} 0 & 1 & 2 \\ 1 & 2 & 1 \\ 2 & 7 & 8 \end{bmatrix}$$

Q2) Rank of the matrix is ____________

$$\begin{bmatrix} 1 & 0 & 2 \\ 2 & 1 & 0 \\ 3 & 2 & 1 \end{bmatrix}$$

Q3) Rank of the matrix is ______________

$$\begin{bmatrix} 2 & 6 & 8 \\ 3 & 7 & 10 \\ 4 & 8 & 12 \\ 5 & 9 & 14 \end{bmatrix}$$

Q4) Rank of the matrix is ________________

$$\begin{bmatrix} 1 & 2 & 3 \\ 2 & 3 & 6 \\ 4 & 5 & 9 \end{bmatrix}$$

Q5) What is the row space of the following matrix ______________

$$\begin{bmatrix} 2 & 4 & 6 & 8 \\ 1 & 3 & 0 & 5 \\ 1 & 1 & 6 & 3 \end{bmatrix}$$

Q6)The projection matrix for v= $[3\ 3\ 3]^T$ is ________________

Q7) the projection of [1,-4,2] along [1,-2,-3] is ________________

<u>Solutions:</u>

1. 2
2. 3
3. 2
4. 3
5. Span of $[1\ 0\ 9\ 2]^T$, $[0\ 1\ -3\ 1]^T$
6. 3x3 matrix with all elements with ⅓
7. 1/14 [3 -6 -9]

Week 3: (GAs)

Q1) Projection of [2,-4,4] along [2,-2,1] is ____________________

Q2) The projection matrix for the matrix $v = [2\ 1\ 3]^T$ is __________________

Q3) Projection of [5,-4,1] along [3,-2,4] is ________________

Q4) Null space of the below matrix is: ________________

$$\begin{bmatrix} 2 & 4 & 6 & 8 \\ 1 & 3 & 0 & 5 \\ 1 & 1 & 6 & 3 \end{bmatrix}$$

Q5) 5 peaches and 6 oranges cost 150 rupees. 10 peaches and 12 oranges cost 300 rupees. Form a matrix out of the given information and its rank is

____

Q6) Which of the following would be the smallest subspace containing the first quadrant of the space R^2 is

A. The first quadrant
B. The first and the third quadrant
C. The first and second quadrant
D. The whole space R^2

Q7) Consider a set of 3 paired observations on ((1,6), (-1,3), (3,15))For the closest line *b* to go through these points, which of the following is the least squares solution ?

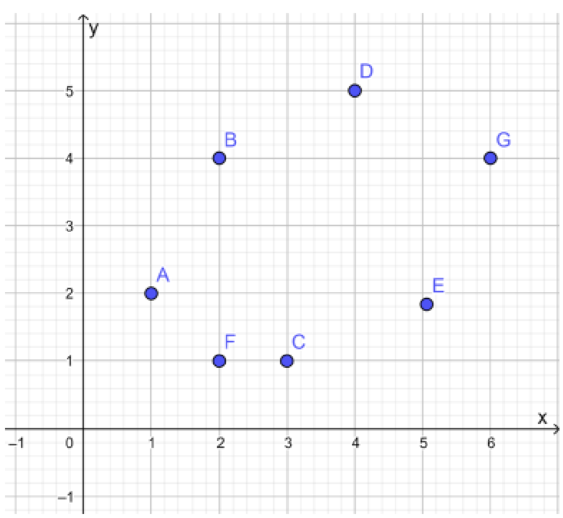
A. (3,4)
B. (4,3)
C. (5,3)
D. (3,5)

Solutions:

1. [32/9 32/9 16/9]
2. $\frac{1}{14}\begin{bmatrix} 4 & 2 & 6 \\ 2 & 1 & 3 \\ 6 & 3 & 9 \end{bmatrix}$
3. $\begin{bmatrix} \frac{18}{29} & \frac{27}{29} & \frac{36}{29} \end{bmatrix}$
4. $\text{Span}\left\{ \begin{bmatrix} -9 \\ 3 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} -2 \\ -1 \\ 0 \\ 1 \end{bmatrix} \right\}$
5. 1
6. D
7. D

Week 3: (Others)

Q1) Consider two lines: (a) A line with zero slope and y-intercept of 2.5. (b) A line with zero slope and y-intercept of 3.5. Which of these lines give the minimal error to fit the data points?



Q2) Find the projection matrix for a = [−1 3 −2 1]^T and use it to obtain the projection of

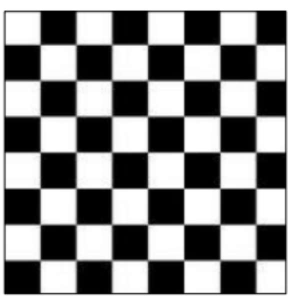
b = [1 0 1 1]^{T}onto a and the length of error vector, e is ____________________

Q3) Can we express b = [4 3 5]^T as a linear combination of the columns of matrix A?

$$A = \begin{bmatrix} 2 & 4 & 6 & 4 \\ 2 & 5 & 7 & 6 \\ 2 & 3 & 5 & 2 \end{bmatrix}$$

Q4) Find the rank of the 8 by 8 checkerboard matrix is __________________

(0 – Black and 1 – White)



Q5) For a matrix A of size m by n,

(a) A and A^T have the same number of pivots.

(b) A and A^T have the same nullspace.

Choose the correct option.

1. (a) – True (b) – False

2. (a) – False (b) – True

3. (a) – True (b) – True

4. (a) – False (b) – False

Q6) What will be the column spaces of these particular matrices?

$$A = \begin{bmatrix} 1 & 2 \\ 0 & 0 \\ 0 & 0 \end{bmatrix} \qquad B = \begin{bmatrix} 1 & 0 \\ 0 & 2 \\ 0 & 0 \end{bmatrix}$$

a) C(A) is a line

b) C(A) is a plane

c) C(B) is a line

d) C(B) is a plane

1. (a) and (c)

2. (a) and (d)

3. (b) and (c)

4. (b) and (d)

Q7) What is the reduced row echelon form of the matrix A _______________?

$$A = \begin{bmatrix} 1 & 0 & -1 & 1 \\ 0 & 1 & 1 & 1 \\ 1 & 1 & 0 & 2 \\ 2 & 3 & 1 & 5 \end{bmatrix}$$

Q8) The length of error vector obtained by projecting u= [-1 4 2]^T onto v=[1 2 3]^T is __________

Q9) A vector x=[a,b] has dot products $x \cdot r$=1 and $x \cdot s$=0 with two given vectors r=[2,−1] and s=[−1,2] a = ____________ b = ______________

Q10) Number of free variables for the below matrix is:_______________

$$A = \begin{bmatrix} 1 & 2 & 2 & 4 & 6 \\ 1 & 2 & 3 & 6 & 9 \\ 0 & 0 & 1 & 2 & 3 \end{bmatrix}$$

Solutions:

1. A
2. 1.652
3. Yes
4. 2
5. 1
6. 2
7. E
8. 3.47
9. 0.33, 0.66
10. 3

Week 4: (Aqs)

Q1) For a linear regression problem $A\theta=Y$, with A, Y as in lecture notes, which of the following is the correct equation of the loss function?

A. $L(\theta)=(½)\,(A\theta-Y)(A\theta-Y)^T$
B. $L(\theta)=(½)\,(A\theta-Y)^T(A\theta-Y)$
C. $L(\theta)=(½)\,(A\theta-Y)^2$

Q2) For $\lambda>0$, $(A+\lambda I)$ is always invertible

A. True
B. False

Q3) For a matrix A with eigenvalue λ, the corresponding eigenvector lies in the

A. null space of A
B. null space of λI
C. null space of $A-\lambda I$
D. null space of $A\lambda$

Q4) Consider A=aI+B where A and B are matrices and a is a scalar. (I is the identity matrix). Which of the following is true?

A. Eigenvectors of A are orthogonal to eigenvectors of B
B. Eigenvectors of A are orthonormal to eigenvectors of B
C. Eigenvectors of A are same as the eigenvectors of B

Q5) For a matrix A having eigenvalue λ, A−λI is

A. an identity matrix
B. a singular matrix
C. a non singular matrix
D. a zero matrix

Q6) All matrices are diagonalizable.

A. *True*
B. *False*

Q7) If the eigenvalues of a matrix are distinct, then its eigenvectors are linearly independent.

A. *True*
B. *False*

Q8) If $S^{-1}AS=\Lambda$, then S is unique.

A. *True*
B. *False*

Q9) eigen value(s) is/are ____________________ & eigen vector(s) is/are__________________ for below matrix:

A= $\begin{bmatrix} 1 & 1 \\ 1 & 0 \end{bmatrix}$

Q10) Suppose that A and P are 2×2 matrices and P is an invertible matrix.

If $P^{-1}AP = \begin{bmatrix} 2 & 1 \\ 0 & 3 \end{bmatrix}$

then the eigenvalues of the matrix A are

A. *1*
B. *2*
C. *3*
D. *4*

Solution:

1. B
2. B
3. C
4. C
5. C
6. B
7. A
8. B
9. 1+sqrt(5)/2, 1-sqrt(5)/2
10. B,c (replace 1 with 0 in the matrix in the question?)

Week 4: (PAs)

Q1) If *P* is a projection matrix, then the eigenvalue corresponding to every nonzero vector lying in the column space of *P* is

A. 0
B. 1
C. -1

Q2) Eigen values of the matrix are ____________

$$A = \begin{bmatrix} 1 & 2 \\ 2 & 3 \end{bmatrix}$$

Q3) If matrix *A* has eigenvalues 1, -2 and 3 then the eigenvalues of A^2 are

A. -1, 2 and -3
B. 1, 4 and 9
C. -1, -4 and -9
D. 1, -2 and 3

Q4) The 90th term of Fibonacci sequence is approximately given by________________

Q5) If x is an eigenvector of a matrix A, then

1. A can shrink x
2. A can stretch x
3. A can change the basis of x
4. A does not change the basis of x

Select all that apply

A. 1, 2, 3 and 4
B. 1, 2 and 3
C. 1, 2 and 4
D. 2, 3 and 4

Q6) If A is a real symmetric $n \times n$ matrix, then which of the following is/are true?

A. A is orthogonally diagonalizable.
B. Eigenvalues of A are real.
C. Eigenvectors corresponding to different eigenvalues are independent.
D. All of these

Q7) Eigen values for the matrix are ____________

$$\begin{bmatrix} 1 & 4 \\ 2 & 3 \end{bmatrix}$$

Q8) eigen vectors for the matrix are _____________

$$A = \begin{bmatrix} 1 & 2 \\ 3 & 2 \end{bmatrix}$$

Q9) Consider following statements regarding a given permutation matrix, *A*

1. 1 is an eigen value of *A*
1. −1 is an eigen value of *A*

Which of the above statements are always true?

A. only 1
B. only 2
C. both 1 and 2
D. neither 1 nor 2

Q10) The best second degree polynomial that fits the data set is ______________

x	y
0	0
1.5	1.5
4	1

Solutions:

1. C
2. 2+sqrt(5), 2-sqrt(5)
3. B
4. 1/sqrt(5) (1+sqrt(5)/2)^90
5. C
6. D
7. 5, -1
8. -
9. A
10. -

Week 4: (GAs)

Q1) If *P* is a projection matrix, then the eigenvalue corresponding to every nonzero vector orthogonal to the column space of *P* is

A. 0
B. 1
C. -1

Q2) Eigen values for the following matrix is/are ___________

$$A=\begin{bmatrix} 1 & 1 \\ 2 & 3 \end{bmatrix}$$

Q3) The 110th term of Fibonacci sequence is approximately given by

____

Q4) Let *A* be an *n*×*n* matrix. Which of the following statements is/are false?

A. If A has *r* non-zero eigenvalues, then rank of *A* is at least *r*.
B. If one of the eigenvalues of *A* are zero, then |A| ≠0.
C. If *x* is an eigenvector of *A*, then so are all the multiples of *x*.
D. If 0 is an eigenvalue of *A*, then *A* cannot be inverted.

Q5) The eigen values of the matrix A^2 is ___________

If $P^{-1}AP = \begin{bmatrix} -1 & 0 & 3 \\ 0 & 3 & 8 \\ 0 & 0 & 4 \end{bmatrix}$

Q6) The best second degree polynomial that fits the data set is ___________

x	y
0	0
1.3	1.5
4	1.2

Solutions:

1. A
2. 2+sqrt(3), 2-sqrt(3)
3. 1/sqrt(5) (1+sqrt(5)/2)^110
4. B
5. 1,9,16
6. -0.316x^2+1.56x

Week 4: (Others)

Q1) The dominant eigen value of the matrix is ___________

$$\begin{bmatrix} 2 & 0 & 4 \\ 0 & 3 & 0 \\ 0 & 1 & 2 \end{bmatrix}$$

Q2) For what value of x, the matrix A, becomes singular?

$$A = \begin{bmatrix} 8 & x & 0 \\ 4 & 0 & 2 \\ 12 & 6 & 0 \end{bmatrix}$$

Q3) The eigenvectors of the below matrix, are written in the form $[1\ a]^T$ and $[1\ b]^T$ what is the value of a+b ___________

$$\begin{bmatrix} 1 & 2 \\ 0 & 2 \end{bmatrix}$$

Q4) What is the value of x, if the below matrix is orthogonal ____________

$$\begin{bmatrix} \frac{3}{5} & \frac{4}{5} \\ x & \frac{3}{5} \end{bmatrix}$$

Q5) Let A be the 2X2 matrix with elements a11 = a12 = a21 = 1 and a22 = -1, then the eigenvalues of A^{18} are? _________________

Q6) The value of p such that the vector $[1\ 2\ 3]^T$ is eigenvector for the below matrix:_______

$$\begin{bmatrix} 4 & 1 & 2 \\ p & 2 & 1 \\ 14 & -4 & 10 \end{bmatrix}$$

Q7) The maximum value of 'a' such that the below matrix, has three linearly independent real eigenvectors is ____________

$$\begin{bmatrix} -3 & 0 & -2 \\ 1 & -1 & 0 \\ 0 & a & -2 \end{bmatrix}$$

Q8) In the matrix equation Px=q, which of the following is a necessary condition for the existence of at least one solution for the unknown vector x.

A) Augmented matrix [Pq] must have the same rank as matrix P.

B) Vector q must have only non zero elements.

C) Matrix P must be singular.

D) Matrix P must be square.

Q9) Consider the following system of equations in three variables x,y and z
2x-y+3z=1
3x-2y+5z=2
-x-4y+z=3
The system of equations has

A. Unique solution
B. Infinitely many solutions
C. No solution
D. none of the above

Q10) For a matrix given below, the eigen value corresponding to eigen vector [101 101]^T is ______

$$\begin{bmatrix} 4 & 2 \\ 2 & 4 \end{bmatrix}$$

Q11) Eigen values for the below matrix are ______________

$$\begin{bmatrix} -1 & 3 & 5 \\ -3 & -1 & 6 \\ 0 & 0 & 3 \end{bmatrix}$$

Solutions:

1. 3
2. 4
3. 0.5
4. -⅘
5. 2^9
6. 17
7. 0.1911
8. -
9. Unique
10. 6
11. -1+3i, -1-3i